Turcan **Tuna**

Senior PhD Candidate · ETH Zurich · Robotics & Spatial 3D Perception

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Research Interest

Main Areas Spatial 3D Perception, Embodied AI, State Estimation & Mapping in the Wild

Education

Ph.D. in Mechanical and Process Engineering (Robotics) — *ETH Zurich*
Robust spatial 3D perception in degenerate environments to enable robot autonomy. — *Feb. 2023 – Present*
Supervisors: Prof. Marco Hutter, Dr. Cesar Cadena

M.Sc. in Robotics, Systems & Control — *ETH Zurich*
Thesis: X-ICP: Localizability-Aware LiDAR Registration for Robust Localization in Extreme Environments. — *Sep. 2019 – May 2022*

B.Sc. in Control & Automation Engineering (Rank: 1st/105) — *Istanbul Technical University*
Thesis: Deep learning frameworks to learn prediction and simulation focused control system models. — *Sep. 2015 – Jun. 2019*

B.Sc. in Mechanical Engineering (Rank: 1st/81) — *Istanbul Technical University*
Thesis: Design and development of FOLLY: A self-foldable and self-deployable quadcopter. — *Sep. 2013 – Jun. 2018*

Publications

* Equal contribution; order within the marked group is a presentation choice.

Journal Articles

GrandTour: A legged robotics dataset in the wild for multi-modal perception and state estimation
Turcan Tuna*, Jonas Frey*, Lanke Frank Tarimo Fu, Katharine Patterson, Tianao Xu, Maurice Fallon, Cesar Cadena, Marco Hutter
The International Journal of Robotics Research (IJRR), OnlineFirst, 1 October 2026

TRaIL-Odom: Tightly Coupled Continuous Time Radar-IMU-LiDAR Odometry With Adaptive Doppler Weighting
Chiyun Noh, **Turcan Tuna**, William Talbot, Marco Hutter, Laurent Kneip, Ayoung Kim
IEEE Robotics and Automation Letters (RA-L), accepted 23 August 2026
Accepted 23 August 2026; assigned to the November 2026 issue, 11(11), 12631–12638. Status recorded as of 22 October 2026.

Holistic Fusion: Task- and Setup-Agnostic Robot Localization and State Estimation With Factor Graphs
Julian Nubert, **Turcan Tuna**, Jonas Frey, Cesar Cadena, Katherine J. Kuchenbecker, Shehryar Khattak, Marco Hutter
IEEE Transactions on Robotics (T-RO) 42, 3366–3387, 2026

CaRLi-V: Camera–RADAR–LiDAR Point-Wise 3-D Velocity Estimation
Landson Guo, Andres M. Diaz Aguilar, William Talbot, **Turcan Tuna**, Marco Hutter, Cesar Cadena
IEEE Robotics and Automation Practice 1, 138–143, 2026

Informed, Constrained, Aligned: A Field Analysis on Degeneracy-Aware Point Cloud Registration in the Wild
Turcan Tuna, Julian Nubert, Patrick Pfreundschuh, Cesar Cadena, Shehryar Khattak, Marco Hutter
IEEE Transactions on Field Robotics 2, 485–515, 2025

Continuous-Time State Estimation Methods in Robotics: A Survey
William Talbot, Julian Nubert, **Turcan Tuna**, Cesar Cadena, Frederike Dümbgen, Jesús Tordesillas, Timothy D. Barfoot, Marco Hutter
IEEE Transactions on Robotics (T-RO) 41, 4975–4999, 2025

X-ICP: Localizability-Aware LiDAR Registration for Robust Localization in Extreme Environments
Turcan Tuna, Julian Nubert, Yoshua Nava, Shehryar Khattak, Marco Hutter
IEEE Transactions on Robotics (T-RO) 40, 452–471, 2024

Scientific exploration of challenging planetary analog environments with a team of legged robots
Philip Arm, Gabriel Waibel, Jan Preisig, **Turcan Tuna**, Ruyi Zhou, Valentin Bickel, Gabriela Ligeza, Takahiro Miki, Florian Kehl, Hendrik Kolvenbach, Marco Hutter
Science Robotics 8(80), eade9548, 2023

Deep learning frameworks to learn prediction and simulation focused control system models
Turcan Tuna, Aykut Beke, Tufan Kumbasar
Applied Intelligence 52(1), 662–679, 2022

Design and development of FOLLY: A self-foldable and self-deployable quadcopter
Turcan Tuna, Salih Ertug Ovur, Etka Gokbel, Tufan Kumbasar
Aerospace Science and Technology 100, 105807, 2020

Conference Proceedings

BIEVR-LIO: Robust LiDAR-Inertial Odometry through Bump-Image-Enhanced Voxel Maps
Patrick Pfreundschuh, **Turcan Tuna**, Cedric Le Gentil, Roland Siegwart, Cesar Cadena, Helen Oleynikova
Robotics: Science and Systems (RSS), 2026

Boxi: Design Decisions in the Context of Algorithmic Performance for Robotics
Turcan Tuna[*], Jonas Frey[*], Lanke Frank Tarimo Fu[*], Cedric Weibel, Katharine Patterson, Benjamin Krummenacher, Matthias Müller, Julian Nubert, Maurice Fallon, Cesar Cadena, Marco Hutter
Robotics: Science and Systems (RSS), 2025

ForestLPR: LiDAR Place Recognition in Forests Attentioning Multiple BEV Density Images
Yanqing Shen, **Turcan Tuna**, Marco Hutter, Cesar Cadena, Nanning Zheng
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 — Highlight

Diffusion Based Robust LiDAR Place Recognition
Benjamin Krummenacher, **Turcan Tuna**[*], Jonas Frey[*], Olga Vysotska, Marco Hutter
IEEE International Conference on Robotics and Automation (ICRA), 2025

LEVA: A High-Mobility Logistic Vehicle with Legged Suspension
Marco Arnold, Lukas Hildebrandt, Kaspar Janssen, Efe Ongan, Pascal Bürge, Ádám Gyula Gábriel, James Kennedy, Rishi Lolla, Quanisha Oppliger, Micha Schaaf, Joseph Church, Michael Fritsche, Victor Klemm, **Turcan Tuna**, Giorgio Valsecchi, Cedric Weibel, Michael Wüthrich, Marco Hutter
IEEE International Conference on Robotics and Automation (ICRA), 2025

Global Path Planning for Autonomous Vehicles in Orchards and Vineyards
Timo Schönegg, **Turcan Tuna**, Fan Yang, Gabriel Waibel, Matias Mattamala, Marco Hutter
International Workshop on Robot Motion and Control (RoMoCo), 2024

FOLLY: A Self Foldable and Self Deployable Autonomous Quadcopter
Turcan Tuna, Salih Ertug Ovur, Etka Gokbel, Tufan Kumbasar
Conference on Control Engineering & Information Technology (CEIT), 2018 — Best Paper Award

Workshop Contributions

Autonomous Forest Inventory with Legged Robots: System Design and Field Deployment
Matías Mattamala, Nived Chebrolu, Benoit Casseau, Leonard Freißmuth, Jonas Frey, **Turcan Tuna**, Marco Hutter, Maurice Fallon
ICRA Workshop on Field Robotics, 2024
Accepted to the ICRA 2024 Workshop on Field Robotics.

Preprints

One-Shot Badminton Shuttle Detection for Mobile Robots
Florentin Dipner, William Talbot, **Turcan Tuna**, Andrei Cramariuc, Marco Hutter
arXiv preprint, 2026; submitted to IEEE Robotics and Automation Practice
Preprint; journal acceptance not verified as of 22 October 2026.

Large-Scale Autonomous Gas Monitoring for Volcanic Environments: A Legged Robot on Mount Etna
Julia Richter, **Turcan Tuna**, Manthan Patel, Takahiro Miki, Devon Higgins, James Fox, Cesar Cadena, Andres Diaz, Marco Hutter
arXiv preprint, 2026; submitted to IEEE Robotics & Automation Magazine
Preprint; journal acceptance not verified as of 22 October 2026.

Correction Notice

Correction to: Deep learning frameworks to learn prediction and simulation focused control system models
Turcan Tuna, Aykut Beke, Tufan Kumbasar
Applied Intelligence 52(1), 680, 2022 — Correction
Funding acknowledgement correction to doi:10.1007/s10489-021-02416-0.

Work and Research Experience

Research Exchange — *MA, USA*
MIT — SPARK Lab — *Sep. 2024 – Dec. 2024*
- Research on spatial 3D mapping with Gaussian splatting for a generalizable robotic mapping pipeline.

Ph.D. Candidate — *Zurich, Switzerland*
ETH Zurich — Robotic Systems Lab — *Feb. 2023 – Present*
- Research on spatial 3D perception in geometrically degenerate environments using multimodal sensors.
- Develop degeneracy-aware point cloud registration methods and analyze mitigation strategies for SLAM.

Scientific Researcher — *Zurich, Switzerland*
ETH Zurich — Robotic Systems Lab — *May 2022 – Feb. 2023*
- Independent research on robust localization and mapping for legged robots.
- Developed X-ICP for localizability-aware LiDAR registration and conducted large-scale evaluation.

Teaching Assistant — Robot Dynamics — *Zurich, Switzerland*
ETH Zurich, Department of Mechanical and Process Engineering — *Sep. 2022 – Jan. 2023*
- Assisted with lectures and exercises for the Robot Dynamics course (4 ECTS).
- Prepared and graded assignments and examinations; supported students during lab sessions.

Perception Engineering Intern — *Zurich, Switzerland*
ANYbotics AG — *Mar. 2021 – Oct. 2021*
- Implemented global initialization and robust localization/mapping for ANYmal quadrupedal robots.
- Integrated LiDAR, camera and inertial sensors for sensor-fusion odometry.
- Conducted field trials in multifloor facilities and on rough terrain.

Student Mechanical Designer — *Zurich, Switzerland*
Autonomous Systems Lab, ETH Zurich — *Sep. 2019 – Mar. 2021*
- Redesigned Super Mega Bot for the DARPA Subterranean Challenge, focusing on robustness and maintainability.
- Performed mechanical design, optimization and finite element analysis; oversaw manufacturing and assembly.

Research Assistant — *Istanbul, Turkey*
Istanbul Technical University — *Sep. 2018 – Jun. 2019*
- Worked on deep learning for nonlinear system identification and time-series prediction.
- Co-authored the journal article on prediction- and simulation-focused models for control systems.

Research Intern — *London, United Kingdom*
Imperial College London — *Jun. 2018 – Sep. 2018*
- Developed algorithms to infer mental states using visual and biosignal data for human–robot interaction.
- Combined EEG, EMG and video data in machine-learning pipelines to estimate human stress levels.

Teaching

2022– ETH Zurich — Robotics Summer School lecturer and teaching assistant.
2023– ETH Zurich — Master's thesis and semester project supervision.
2022–23 ETH Zurich — Robot Dynamics teaching assistant.

Awards

2025 CVPR Highlight — ForestLPR: LiDAR Place Recognition in Forests Attentioning Multiple BEV Density Images.
2024 RoMoCo Young Author Best Paper Award — Global Path Planning for Autonomous Vehicles in Orchards and Vineyards; Timo Schönegg received the award for the co-authored paper.
2021 ISIF Silver Medal — with Etka Gökbel and Salih Ertuğ Övür, for a patent registered in Turkey.
2018 CEIT Best Paper Award — FOLLY: A Self Foldable and Self Deployable Autonomous Quadcopter.
2019 Top Graduate — Control & Automation Engineering, Istanbul Technical University (1st/105).
2018 Top Graduate — Mechanical Engineering, Istanbul Technical University (1st/81).

Service and Invited Talks

2026 IROS workshop organizer — Data in Field Robotics: From State Estimation to Navigation, 1 October.
2025 INTERGEO talk — GrandTour and Boxi, 9 October.
2025 Scientific keynote, FZI RimA Summer School — Towards Walking Robots in Domestic Environments, 13 August.